

DEVA ANAND

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Education

MS, Computer Science , University of Massachusetts Amherst	2025–2027
<i>Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering</i>	
BE, Computer Science and Engineering , Vels Institute of Science, Technology and Advanced Studies	2019–2023

Professional Experience

TherAlign Health , Founding Software Engineer Intern <i>Remote</i>	Jul 2026 – Present
- Founding engineer building a provider-facing clinical decision support app (Next.js + Firebase/GCP) where an LLM-backed pipeline generates guideline-grounded medication recommendations inside the EHR prescribing workflow. - Extending a Firebase Cloud Functions backend where Gemini is constrained to synthesis while the backend deterministically controls eligibility and evidence labels, a safety-guardrail pattern that keeps an autonomous pipeline grounded and verifiable rather than free-form.	
Culture & Morality Lab, UMass Amherst , Lab Assistant <i>Amherst, MA</i>	Jul 2026 – Present
- Sole engineer productionizing the lab's published CCR analysis method into a self-serve research data platform (Python, FastAPI, sentence-transformers, React) that runs ingest-to-scores end to end, owning architecture and all technical decisions. - Designed versioned model/construct registries, per-run reproducibility exports, and a structured data-quality warning system, plus 40 hermetic tests in CI, quality and guardrail infrastructure so automated analyses stay trustworthy and repeatable.	
Wysa , Backend Engineer <i>Bengaluru, India</i>	Jul 2023 – Jul 2025
- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team that automated repeatable labeling work, cutting manual effort by 100+ hours/month and lifting throughput by 60%. - Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for 8+ UK clinical clients, processing 10,000+ monthly triage submissions via Mayden iaptus REST integrations; contributed to \$340K+ in revenue. - Acted as primary technical point of contact for the eTriage backend, shipping authentication, rate limiting, and audit logging, and running structured knowledge transfer that onboarded a second engineer to on-call readiness in ~6 weeks.	

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub Demo)	Python, FastAPI, OpenAI / Groq, SQLite
Built an agentic system that acts rather than chats: it plans typed retrieval, runs web search and reranking, extracts entities deterministically with LLM fallback, and verifies its own output, returning cell-level provenance; hardened with 150+ tests and provider routing/fallback.	
AutoEval – Autonomous Evaluation Agent (GitHub)	Python, IR evaluation, Git automation
Built an autonomous evaluation agent with a holdout-heavy fitness objective and anti-Goodharting guardrails (schema checks, duplicate rejection, read-only holdout protection), committing accepted updates to Git with report cards for an auditable agent-evaluation loop.	
cf-ai-research-scout – Edge-Native AI Agent (GitHub)	TypeScript, Cloudflare Workers AI, Durable Objects, D1
Built an edge-native AI agent whose multi-step research tasks survive process restarts and individual tool failures via durable workflow orchestration (WorkflowEntrypoint + Durable Objects), with agent state persisted server-side in D1.	

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: Python, TypeScript, JavaScript, SQL, C++, Bash
LLM & Agents: agentic systems, tool-use / self-verification, provider routing / fallback, prompt iteration, RAG, LlamaIndex, evaluation guardrails
ML & Eval: PyTorch, sentence-transformers, scikit-learn, NumPy, pandas, Hugging Face Transformers, model evaluation
Backend & APIs: FastAPI, Node.js, Express, REST APIs, async services, Docker
Cloud & DevOps: AWS, GCP, Azure, Git, GitHub Actions, Linux, CI/CD

Open Source & Community

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.